HANDS-ON LAB WORKBOOK

Google Cloud Platform

Practical Cloud Exercises

A step-by-step lab manual for engineering students. Build, connect, and verify real resources across the five core service families.

AUDIENCE	DURATION	FORMAT
Engineering Students	~2 Hours	5 Guided Labs

Lecture by **Imran Sayyed**

GETTING STARTED · SETUP

What You Need Before You Start

A short checklist to make sure your lab environment is ready.

GOOGLE ACCOUNT

A Google login with Cloud Console access.

WEB BROWSER

Chrome or any modern browser; no install required.

BILLING / FREE TIER

A billing account or free-trial credits enabled.

A PROJECT

A dedicated project to hold all lab resources.

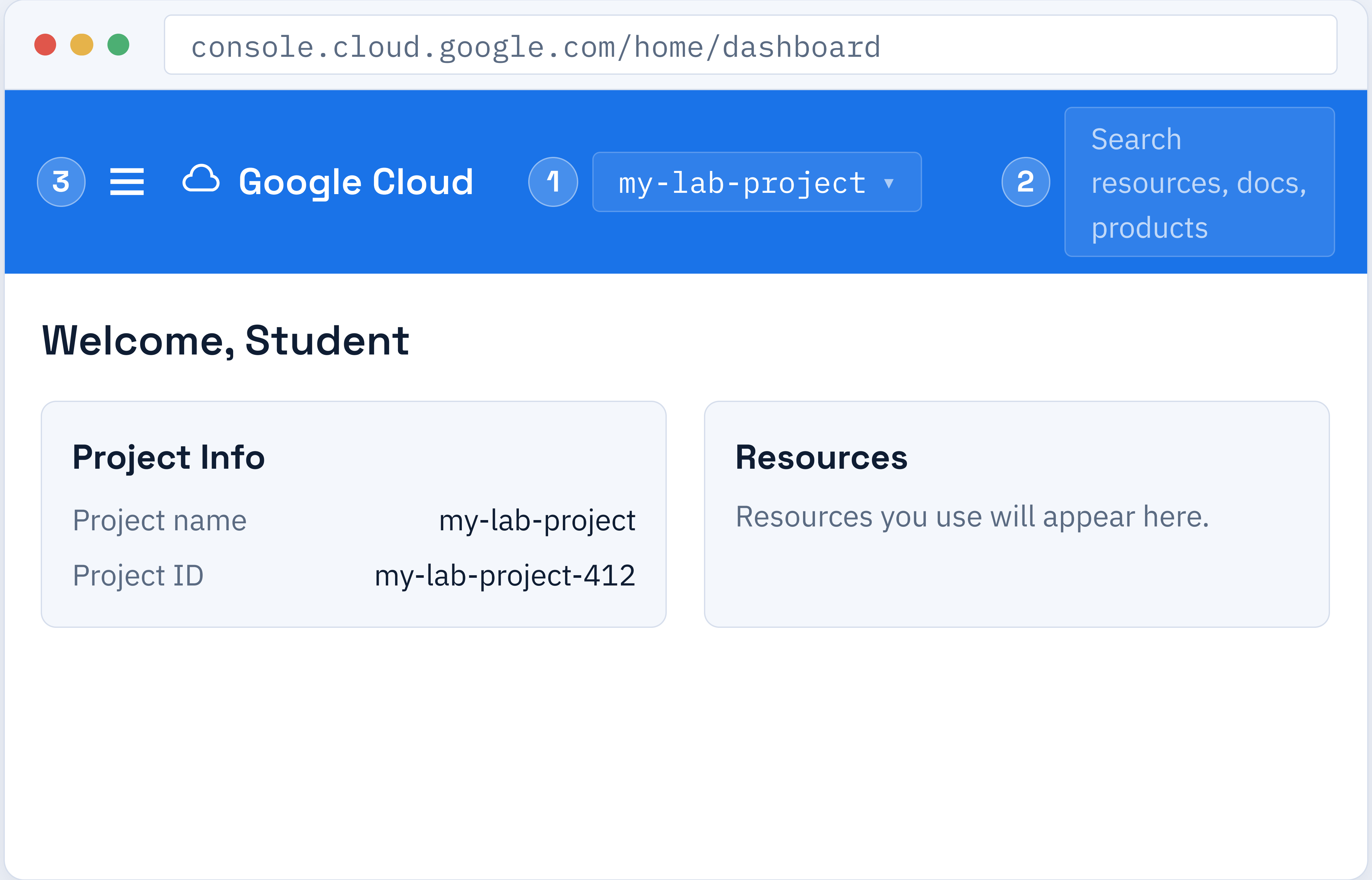
GETTING STARTED · CONSOLE TOUR

Touring the Cloud Console

KEY AREAS

1. **Project picker** — top bar dropdown.
2. **Search bar** — jump to any service.
3. **Navigation menu** — the hamburger at top-left.

✓ **Expected result:** you can identify the project picker, search, and the navigation menu.



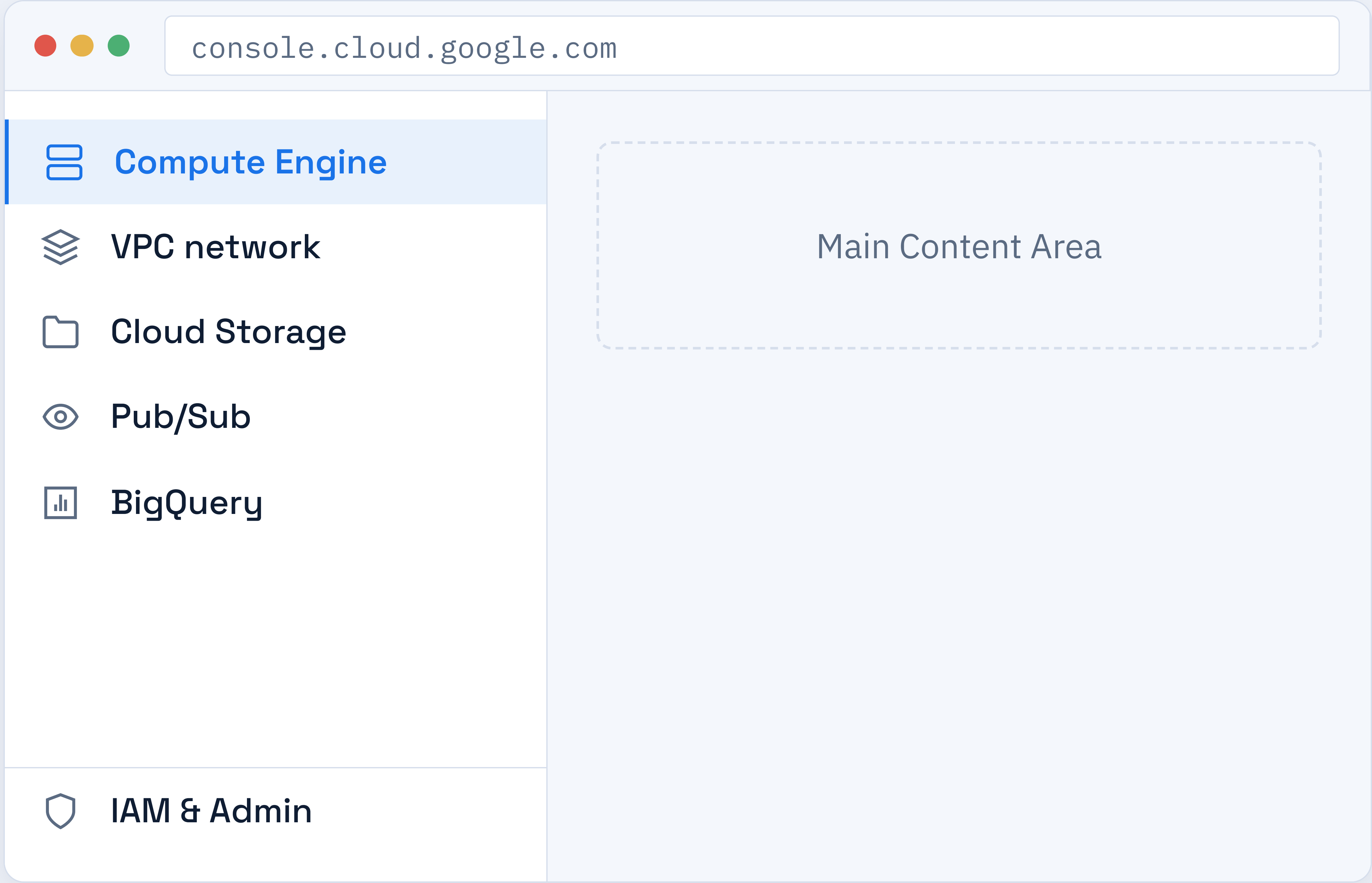
GETTING STARTED · NAVIGATION

Finding Services in the Menu

STEPS

1. Click the **navigation menu** (☰) top-left.
2. Scroll to a product group such as **Compute** or **Storage**.
3. Pin frequently used services for quick access.

✓ **Expected result:** the navigation menu opens and lists product groups.



How Projects Organize Your Work

Every resource you create lives inside a project.

PROJECT

A container for resources, billing, and permissions.

PROJECT ID

A globally unique identifier you cannot change later.

ISOLATION

Resources in one project are separate from another.

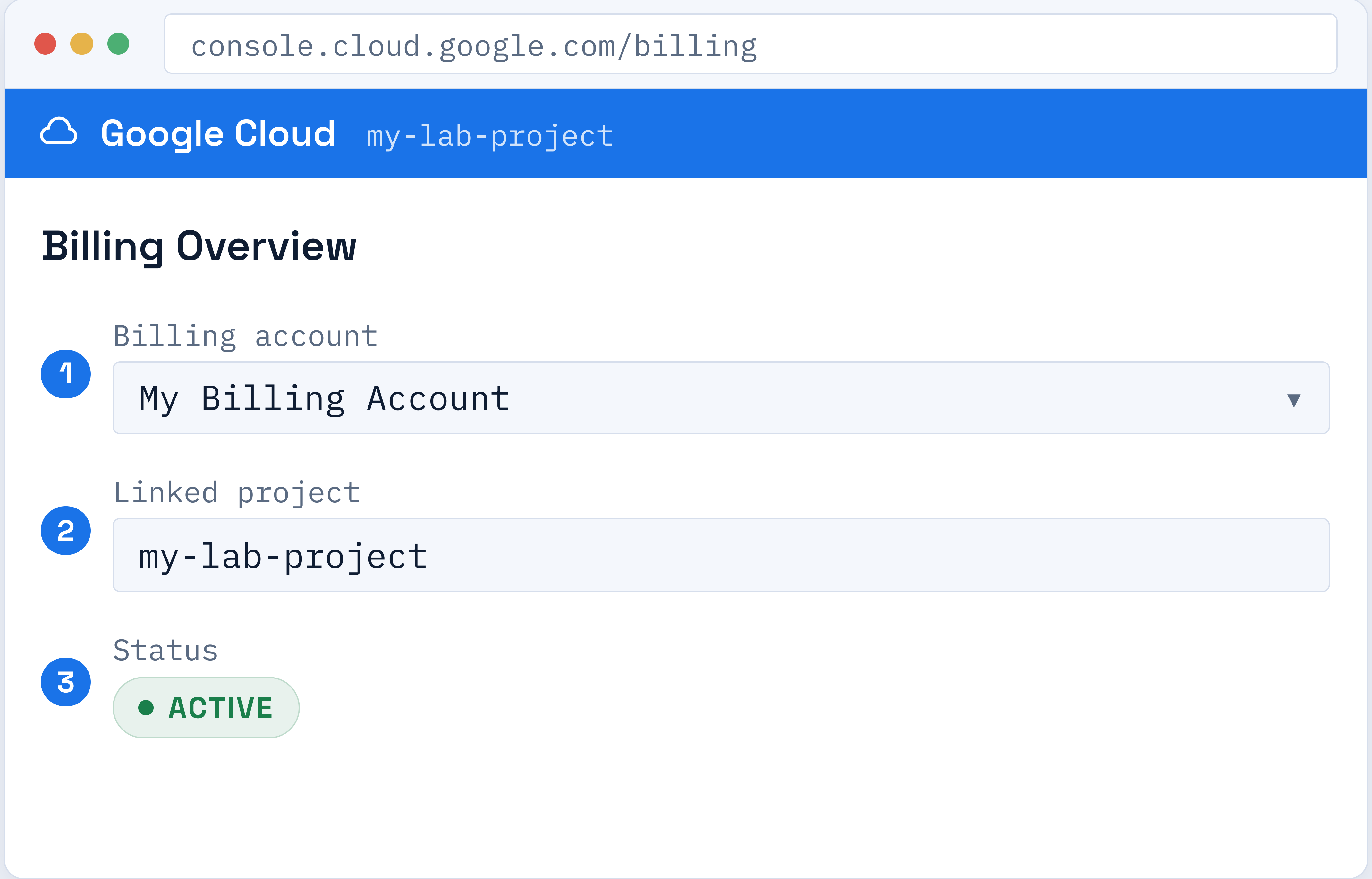
Organization → **Project** → Resources

GETTING STARTED · BILLING

Confirm Billing Is Linked

STEPS

1. Open **Billing** from the navigation menu.
2. Check that a billing account is linked to your project.
3. Confirm the status shows **Active**.



✓ **Expected result:** billing shows Active and the project is linked.

COMPUTE ENGINE LAB · OBJECTIVE

Create & Connect to a Linux VM

Provision a virtual machine on Compute Engine, open a firewall path, and confirm you can reach it over SSH.

OBJECTIVE

What you will accomplish in this lab

STEPS

Numbered console actions to follow in order

EXPECTED RESULT

How to confirm the step worked

COMMON ERRORS

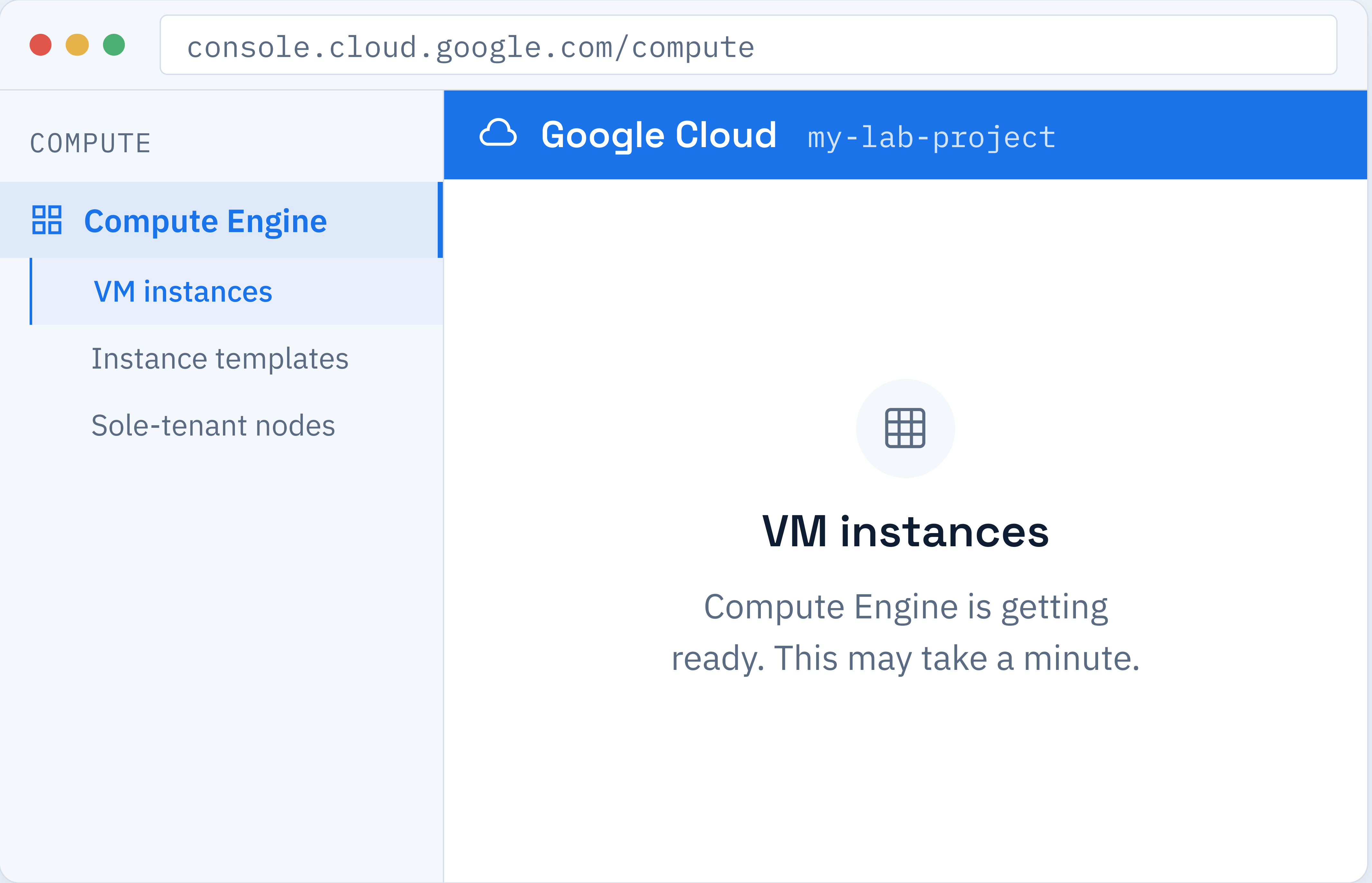
Pitfalls to watch for and how to fix them

Open Compute Engine

STEPS

1. Click the navigation menu (☰).
2. Under **Compute**, select **Compute Engine**.
3. Open the **VM instances** page.

✓ **Expected result:** the VM instances page loads (it may be empty at first).



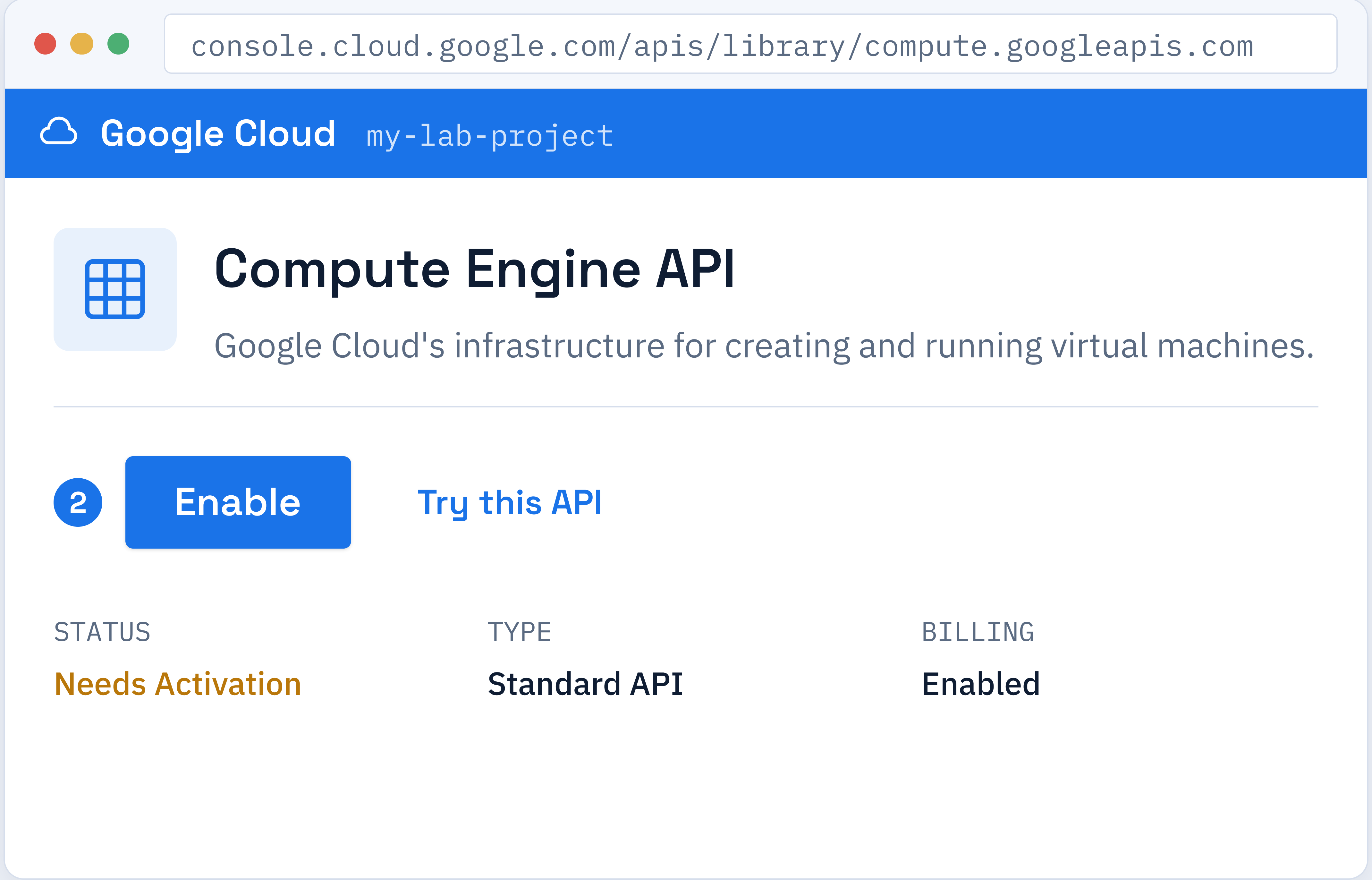
COMPUTE ENGINE LAB · STEP 2

Enable the Compute Engine API

STEPS

1. The first visit prompts to enable the API.
2. Click **Enable**.
3. Wait for activation to finish.

✓ **Expected result:** the API shows Enabled and the instances page is ready.

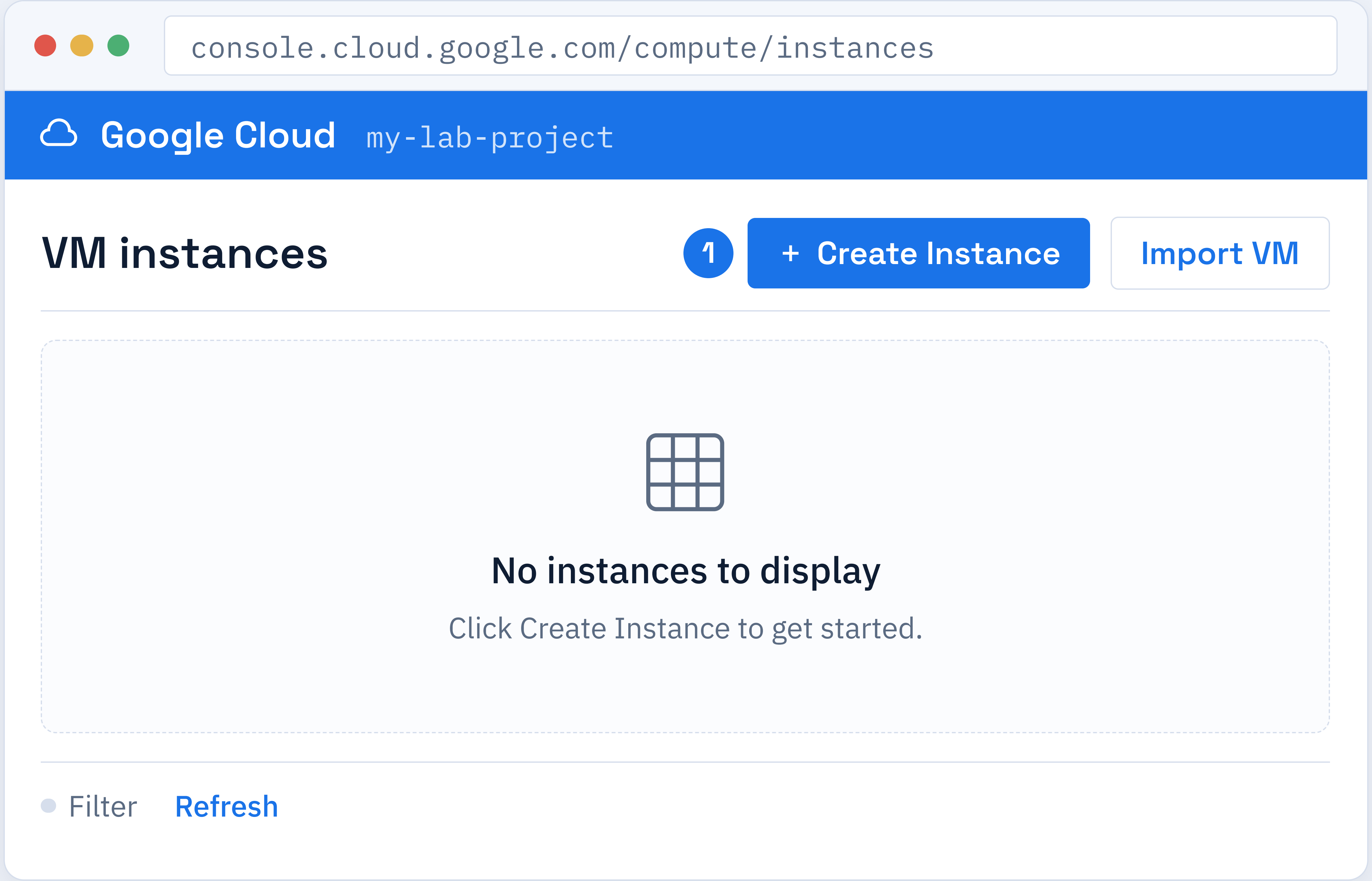


COMPUTE ENGINE LAB · STEP 3

Start a New Instance

STEPS

1. On the VM instances page, click **Create Instance**.
2. The instance creation form opens.
3. You will fill it in over the next steps.



✓ **Expected result:** the Create an instance form is displayed.

COMPUTE ENGINE LAB · STEP 4

Configure the VM Instance

STEPS

1. Enter a **Name** such as `lab-vm-01`.
2. Pick a **Region** close to you, e.g. `us-central1`.
3. Select a **Zone** within that region.

✓ **Expected result:** name and location fields are filled and the form is ready for machine type.

COMPUTE ENGINE LAB · STEP 5

Choose a Machine Type

STEPS

1. Open the **Machine type** dropdown.
2. For labs, pick a small, low-cost type.
3. `e2-micro` is free-tier eligible.

✓ **Expected result:** a machine type is selected (e2-micro recommended).

console.cloud.google.com/compute/instancesAdd

Google Cloud my-lab-project

Machine configuration

✓	`e2-micro` 2 vCPU (shared), 1 GB memory	Free-tier eligible
	`e2-small` 2 vCPU, 2 GB memory	~$12.00 / month
	`e2-medium` 2 vCPU, 4 GB memory	~$24.00 / month

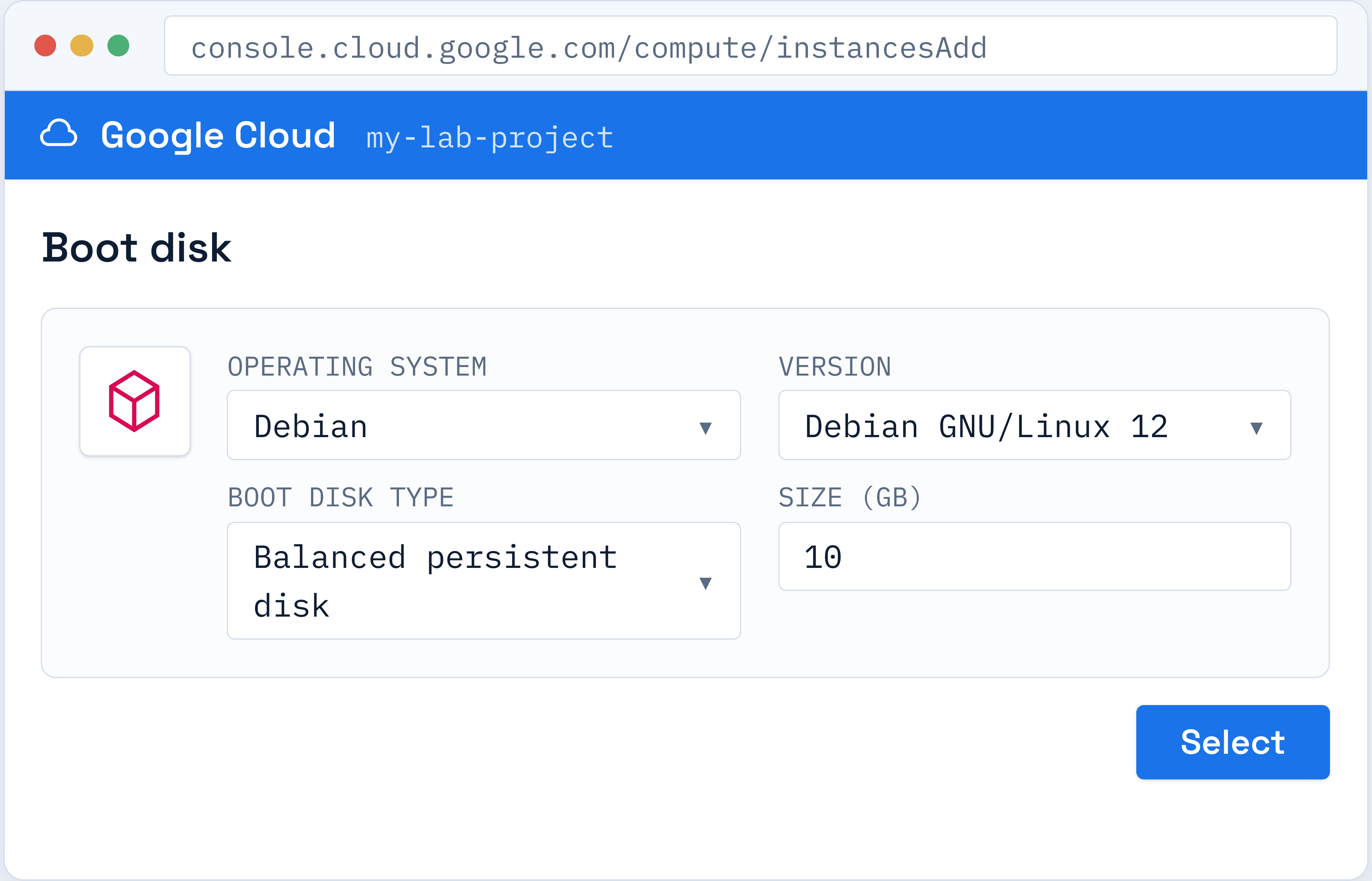
ⓘ Cost estimates are based on 730 hours of usage per month. Actual costs will be lower for short-lived lab VMs.

COMPUTE ENGINE LAB · STEP 6

Pick the Boot Disk Image

STEPS

1. Open the **Boot disk** section.
2. Choose an OS image, e.g. **Debian** or **Ubuntu**.
3. Keep the default disk size for labs.



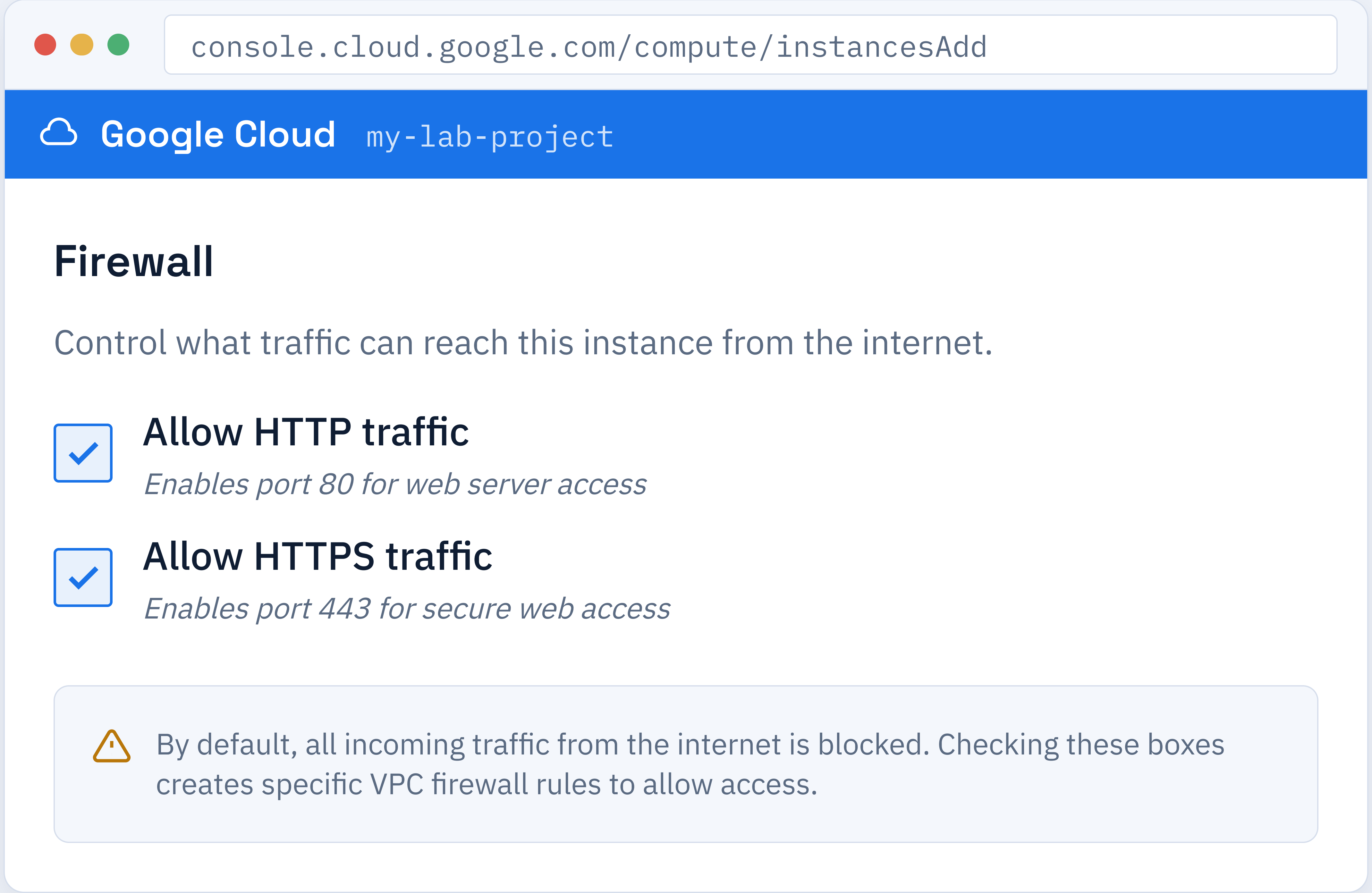
✓ **Expected result:** the boot disk shows your chosen Linux image.

COMPUTE ENGINE LAB · STEP 7

Set Firewall Options

STEPS

1. Find the **Firewall** section in the form.
2. Check **Allow HTTP traffic**.
3. Check **Allow HTTPS traffic** if needed.



✓ **Expected result:** the HTTP (and HTTPS) firewall boxes are checked.

COMPUTE ENGINE LAB · STEP 8

Review and Create

STEPS

1. Review the instance summary.
2. Check name, region, machine type.
3. Click **Create**.

✓ **Expected result:** the VM begins provisioning and appears in the instances list.

console.cloud.google.com/compute/instancesAdd

Google Cloud my-lab-project

Summary

NAME lab-vm-01	IMAGE Debian GNU/Linux 12
REGION us-central1	FIREWALL Allow HTTP, HTTPS
MACHINE TYPE e2-micro (2 vCPU, 1 GB)	ESTIMATED COST Free-tier eligible

3 **Create** Cancel

COMPUTE ENGINE LAB · STEP 9

Connect Over SSH

STEPS

1. In the instances list, find your VM.
2. Click the **SSH** button in its row.
3. A browser terminal opens a session.

✓ **Expected result:** a terminal opens and you see a shell prompt on the VM.

COMPUTE ENGINE LAB · STEP 10

Verify the VM Is Running

STEPS

1. Return to the VM instances list.
2. Confirm the status is **Running** (green).
3. Note the **external IP** address.

✓ **Expected result:** the VM shows status Running with an external IP assigned.

console.cloud.google.com/compute/instances

VM instances

Filter: my-lab-project

NAME	ZONE	STATUS	EXTERNAL IP
lab-vm-01	us-central1-a	Running	34.31.25.114

VM Successfully Deployed

Your instance is live and ready for traffic on the external IP shown above.

COMPUTE ENGINE LAB · COMMON ERRORS

When the VM Won't Cooperate

⚠ Quota exceeded

Free-tier projects limit CPUs per region.

Fix: choose a smaller machine type or a different region.

⚠ API not enabled

The Compute Engine API must be on first.

Fix: open APIs & Services and click Enable.

⚠ SSH connection refused

Firewall blocks port 22 or the VM is still booting.

Fix: allow port 22 and wait for status Running.

⚠ Billing not linked

Resources cannot be created without billing.

Fix: attach a billing account to the project.

NETWORKING LAB · OVERVIEW

How Cloud Networking Works

A few building blocks control how your VM is reached and how it communicates with the world.

VPC NETWORK
A private virtual network for your resources.

SUBNET
A regional IP range inside the VPC.

IP ADDRESSES
Internal for private traffic, external for the internet.

FIREWALL RULES
Allow or deny traffic by port and protocol.

NETWORKING LAB · CONCEPT

External IP Addresses

An external IP makes a VM reachable from the internet. There are two main types to consider.

EPHEMERAL

Assigned automatically, changes when the instance is stopped and restarted.

STATIC

Reserved and fixed. The IP stays the same even if the VM is deleted or moved.

NETWORKING LAB · CONCEPT

Internal IP Addresses

Internal IPs let resources talk privately inside the VPC without exposing traffic to the internet.

PRIVATE RANGE

From the subnet's internal IP range (e.g., 10.128.0.0/20), not internet-routable.

VM-TO-VM

Used for low-latency traffic between instances in the same VPC or peered networks.

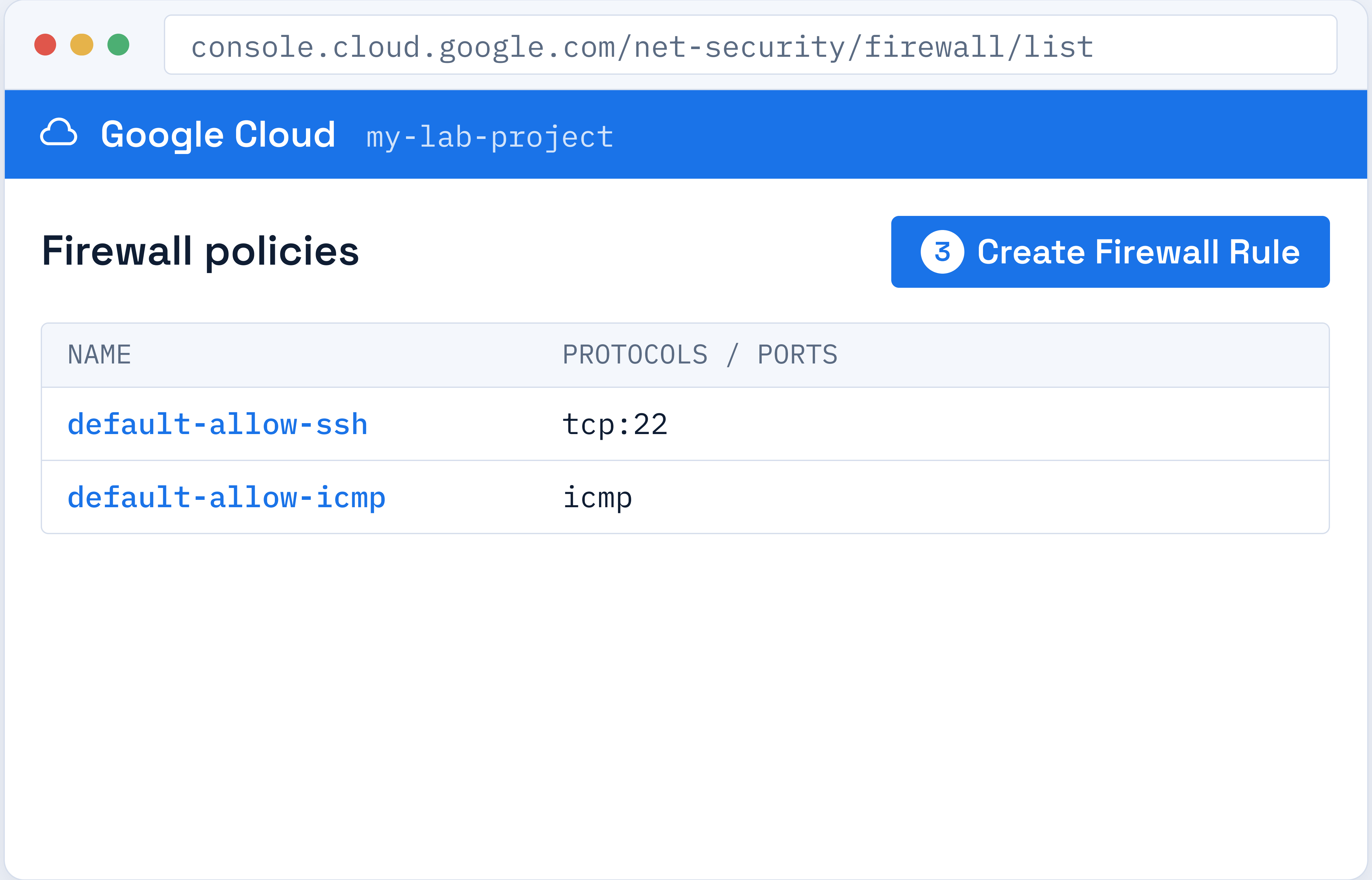
NETWORKING LAB · STEP 1

Open the Firewall Rules Page

STEPS

1. In the navigation menu, go to **VPC network** → **Firewall**.
2. Review the **default rules** already present in the list.
3. Click the **Create Firewall Rule** button at the top.

✓ **Expected result:** the firewall rules list is open and ready for a new rule.

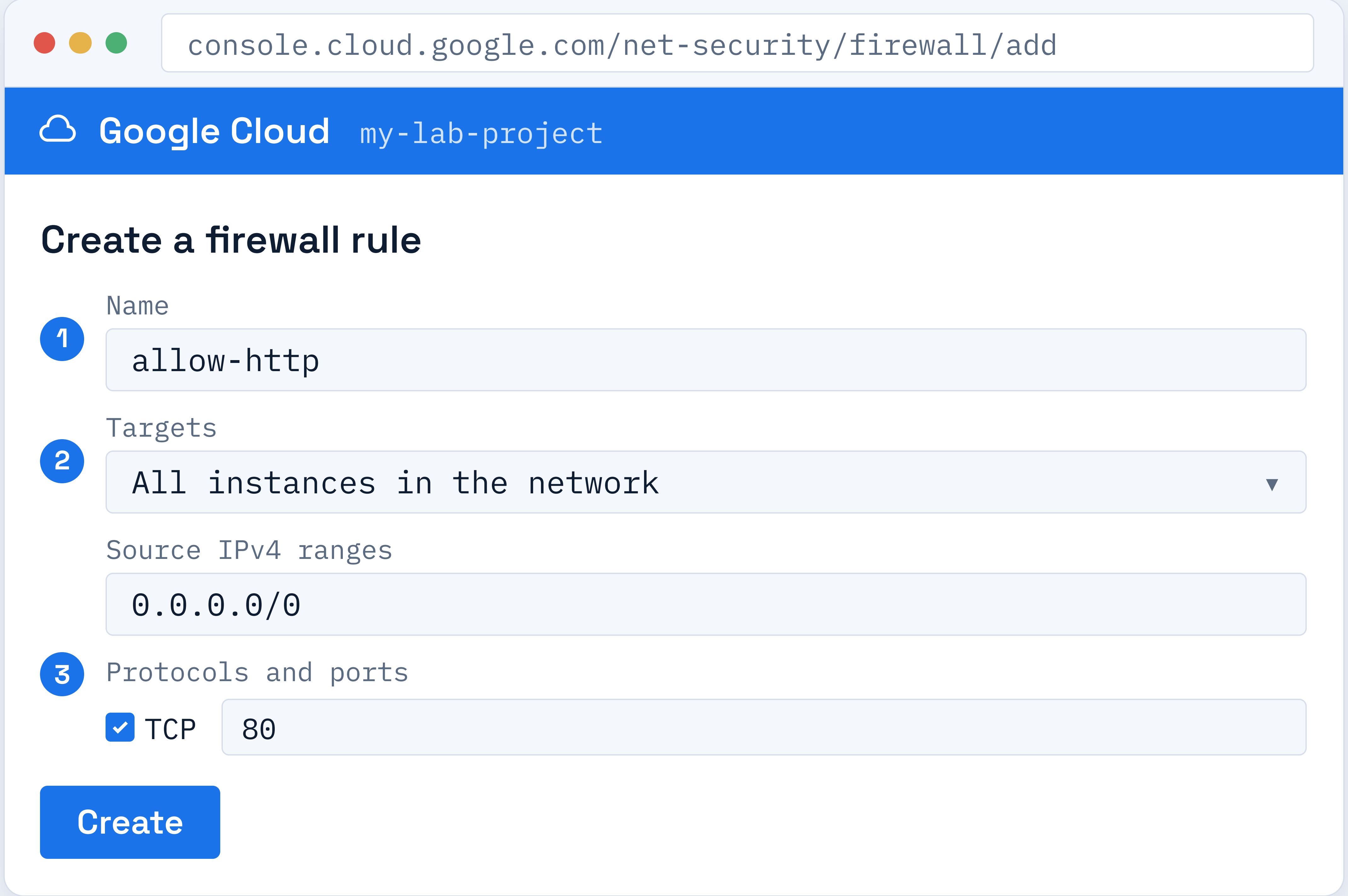


NETWORKING LAB · STEP 2

Allow HTTP Traffic

STEPS

1. Name the rule **`allow-http`**.
2. Set **Targets** to "All instances in the network" and **Source ranges** to `0.0.0.0/0`.
3. In **Protocols and ports**, check **TCP** and enter 80.



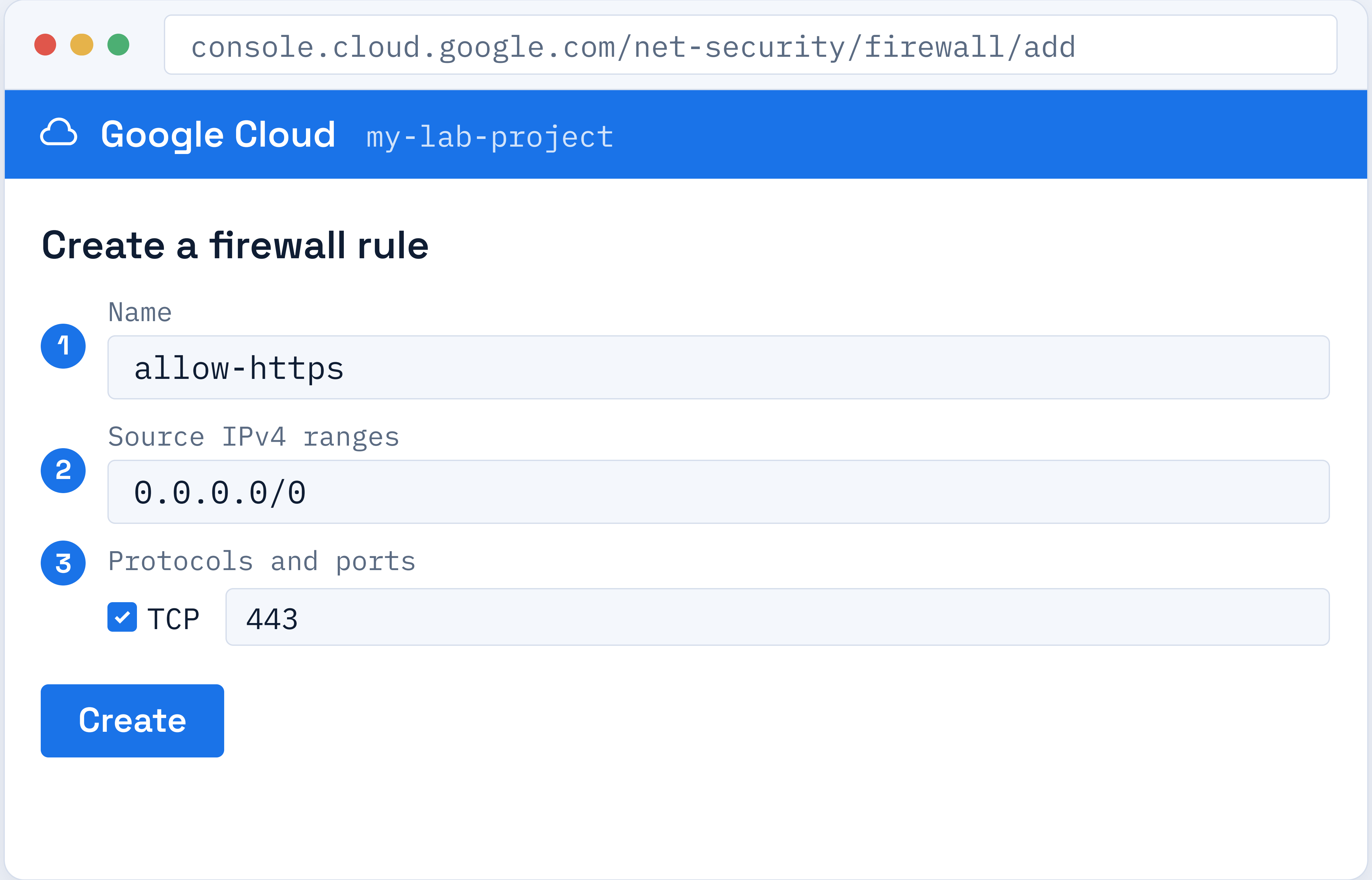
✓ **Expected result:** a rule allowing incoming traffic on port 80 is created.

NETWORKING LAB · STEP 3

Allow HTTPS Traffic

STEPS

1. Create another rule named **`allow-https`**.
2. Set **Source ranges** to `0.0.0.0/0` for universal access.
3. In **Protocols and ports**, enter 443 for TCP.



✓ **Expected result:** a rule allowing incoming secure traffic on port 443 is created.

NETWORKING LAB · COMMON ERRORS

Networking Gotchas

⚠ Forgot the firewall rule

Traffic is blocked by default even if the service is running.

Fix: create an allow rule for the specific port you need.

⚠ Wrong source range

Rule scoped too narrowly (e.g. only your home IP).

Fix: use 0.0.0.0/0 for public access during labs.

⚠ No external IP

VM created without one cannot be reached from the internet.

Fix: assign an external IP (ephemeral or static) to the instance.

⚠ Port mismatch

Service runs on a different port than the firewall allows.

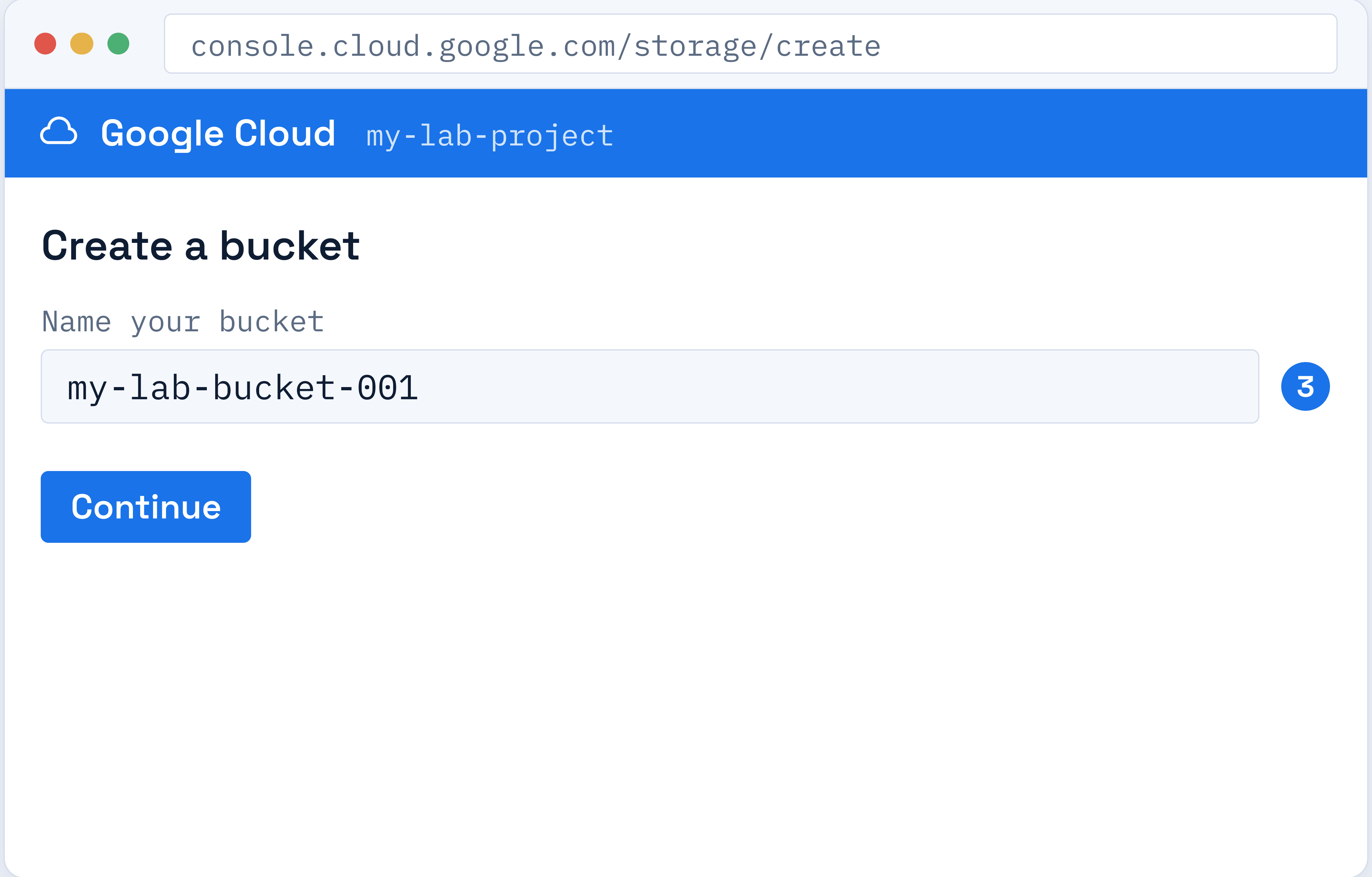
Fix: open the exact port your application is listening on.

CLOUD STORAGE LAB · STEP 1

Create a Storage Bucket

STEPS

1. Open **Cloud Storage → Buckets**.
2. Click **Create**.
3. Enter a globally unique bucket name.



✓ **Expected result:** the bucket creation form opens with your name accepted.

CLOUD STORAGE LAB · CONCEPT

Bucket Naming Rules

Buckets follow strict naming conventions enforced across the entire global service.

GLOBALLY UNIQUE

No two buckets share a name across all of GCP.

LOWERCASE

Letters, numbers, dashes; no uppercase.

LENGTH

3 to 63 characters in total length.

NO SPACES

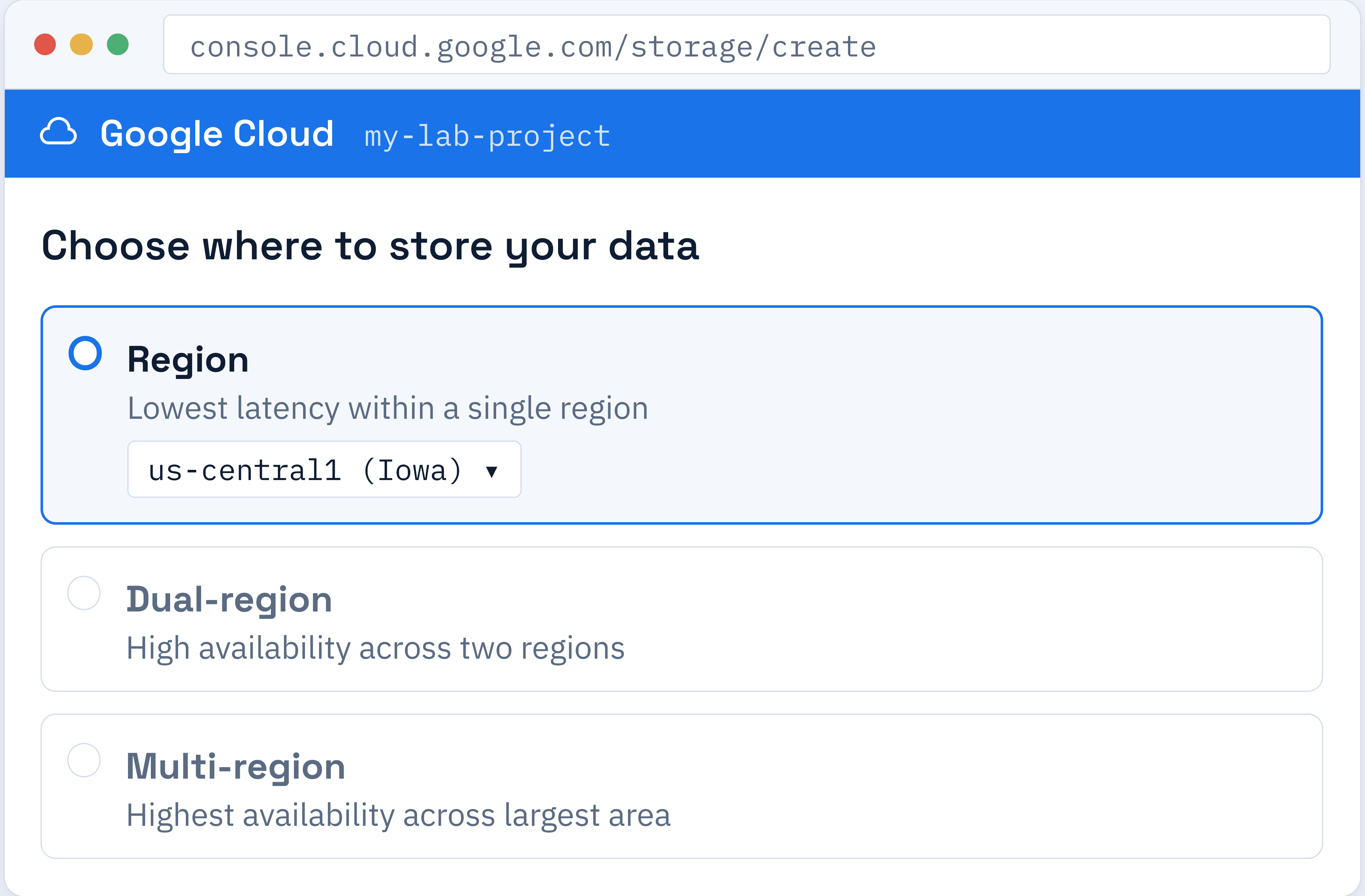
Use dashes, not spaces or special characters.

CLOUD STORAGE LAB · STEP 2

Choose Where Data Lives

STEPS

1. Pick a **Location type**.
2. **Region** keeps data in one place for lowest cost.
3. **Multi-region** spreads data for high availability.



✓ **Expected result:** a location is selected (Region recommended for labs).

CLOUD STORAGE LAB · CONCEPT

Pick a Storage Class

Storage classes balance data availability and performance with cost.

STANDARD

Frequent access, hot data, highest performance.

NEARLINE

Access about once a month, lower storage cost.

COLDLINE

Access about once a quarter, deep storage discount.

ARCHIVE

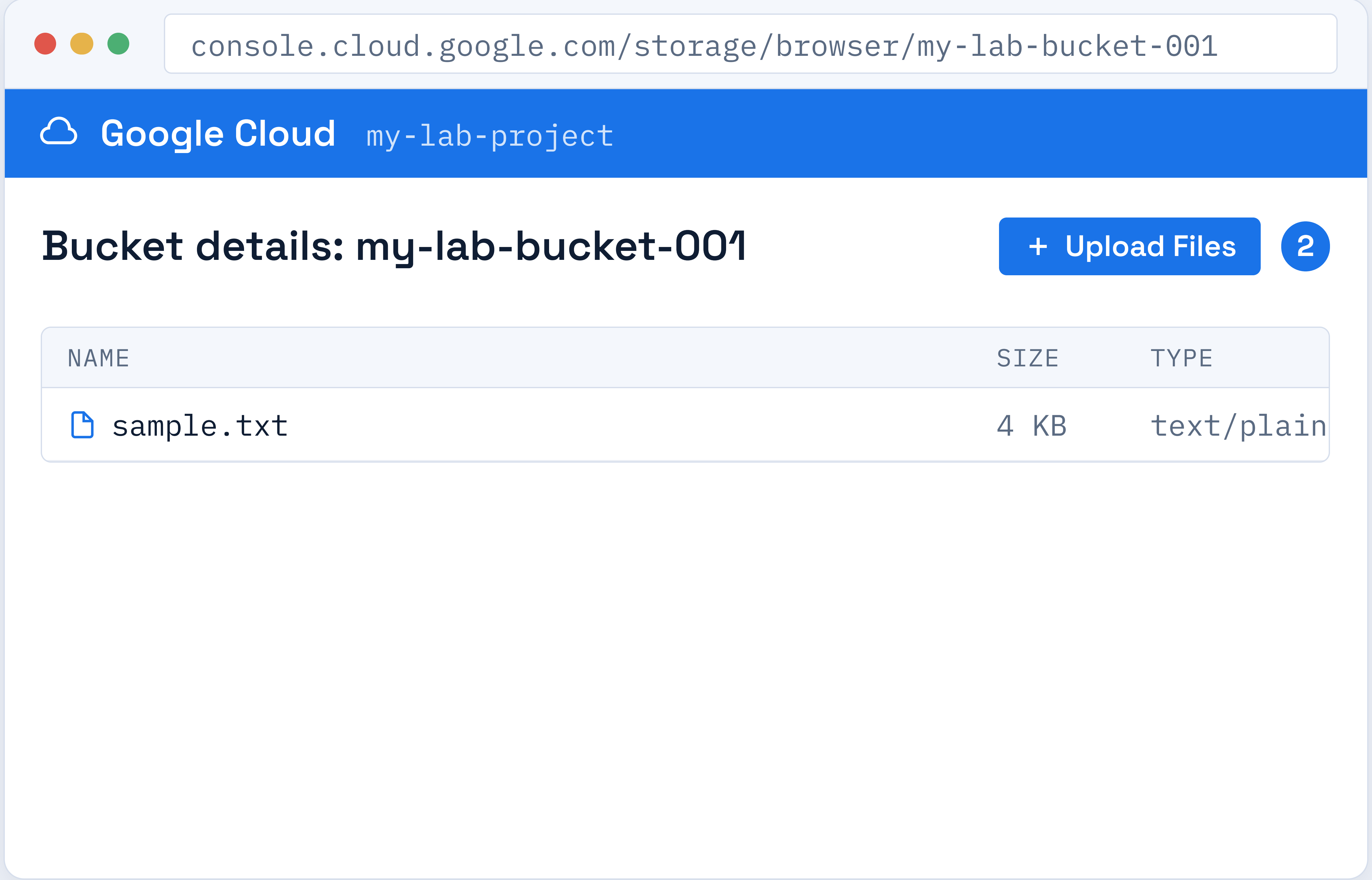
Rare access (once a year), lowest storage cost.

CLOUD STORAGE LAB · STEP 3

Upload an Object

STEPS

1. Open your bucket from the list.
2. Click **Upload Files**.
3. Select a file from your local computer.



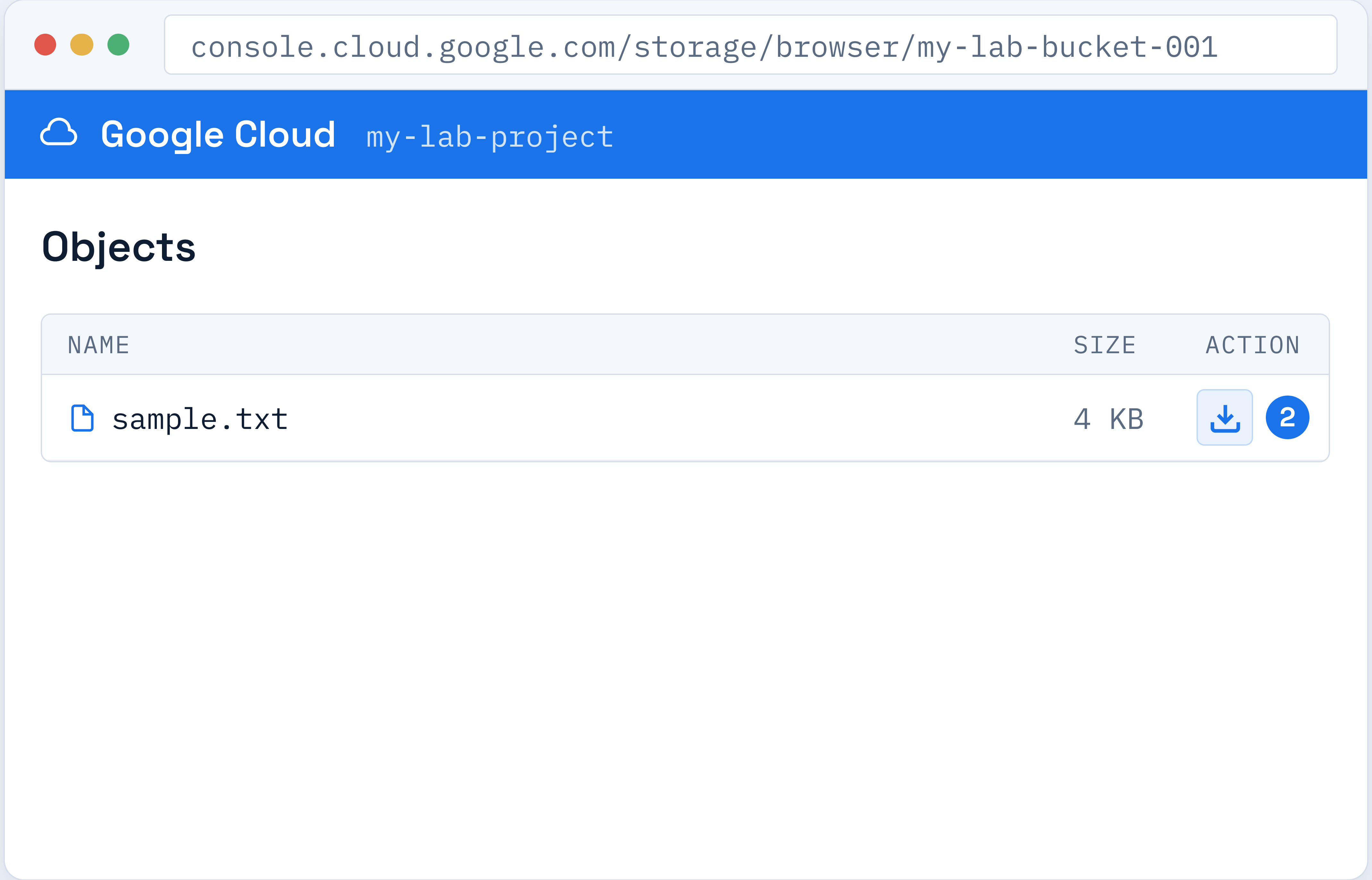
✓ **Expected result:** the file appears in the bucket's object list.

CLOUD STORAGE LAB · STEP 4

Download an Object

STEPS

1. Find the object in the bucket's list.
2. Click the **download icon** in its row.
3. The file saves locally to your device.



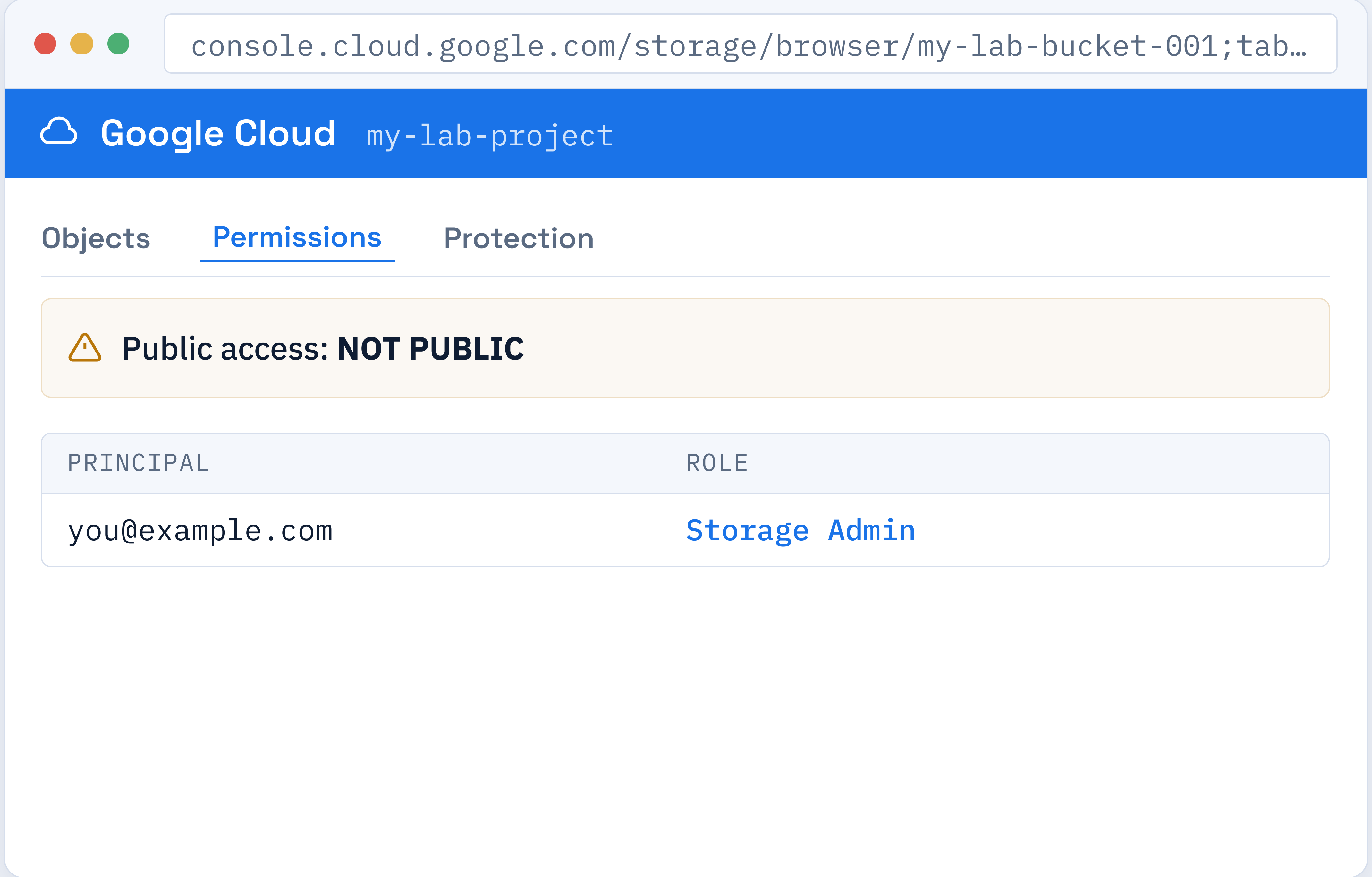
✓ **Expected result:** the object downloads to your computer.

CLOUD STORAGE LAB · STEP 5

Control Who Can Access

STEPS

1. Open the **Permissions** tab.
2. Review IAM roles assigned to the bucket.
3. Avoid making data public unless required.



✓ **Expected result:** you can see the principals and roles on the bucket.

CLOUD STORAGE LAB · COMMON ERRORS

Storage Pitfalls

⚠ Bucket name taken

Bucket names must be globally unique across all of GCP.

Fix: add a unique suffix such as your name or a random number.

⚠ Invalid name

Names can only contain lowercase letters, numbers, and dashes.

Fix: use lowercase letters and dashes only; avoid spaces.

⚠ Access denied

The principal is missing the required IAM permission for the bucket.

Fix: grant the correct Storage Object Viewer or Admin role.

⚠ Accidentally public

The bucket has been granted read access to "allUsers".

Fix: remove the public access entry from the permissions list.

PUB/SUB LAB · OVERVIEW

The Messaging Pipeline

Pub/Sub decouples senders from receivers.

Publisher

Sends messages

→

Topic

Receives and stores messages

→

Subscriber

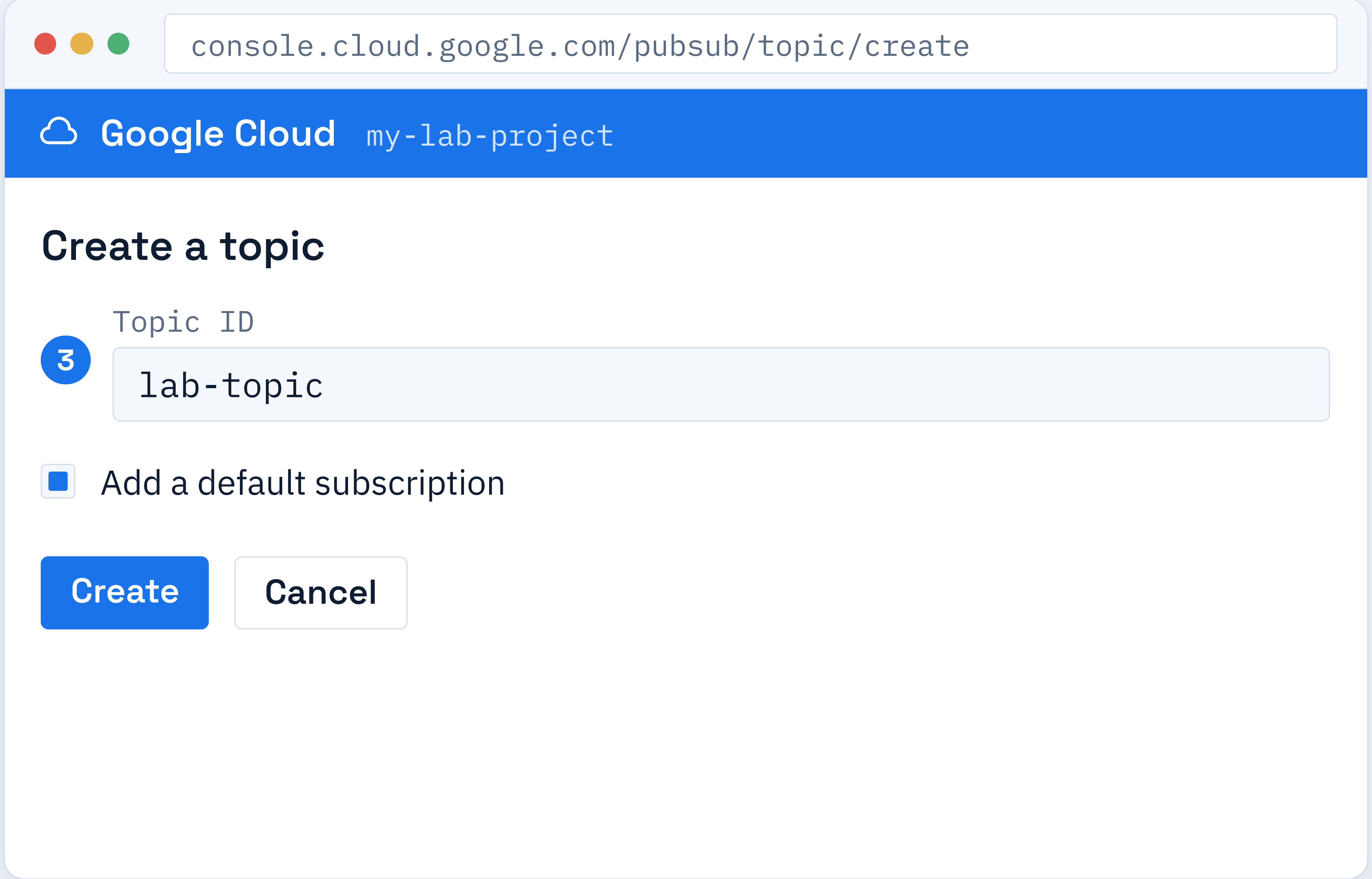
Pulls and processes messages

PUB/SUB LAB · STEP 1

Create a Topic

STEPS

1. Open **Pub/Sub → Topics**.
2. Click **Create Topic**.
3. Enter a **Topic ID**, e.g. `lab-topic`.



✓ **Expected result:** the topic appears in the topics list.

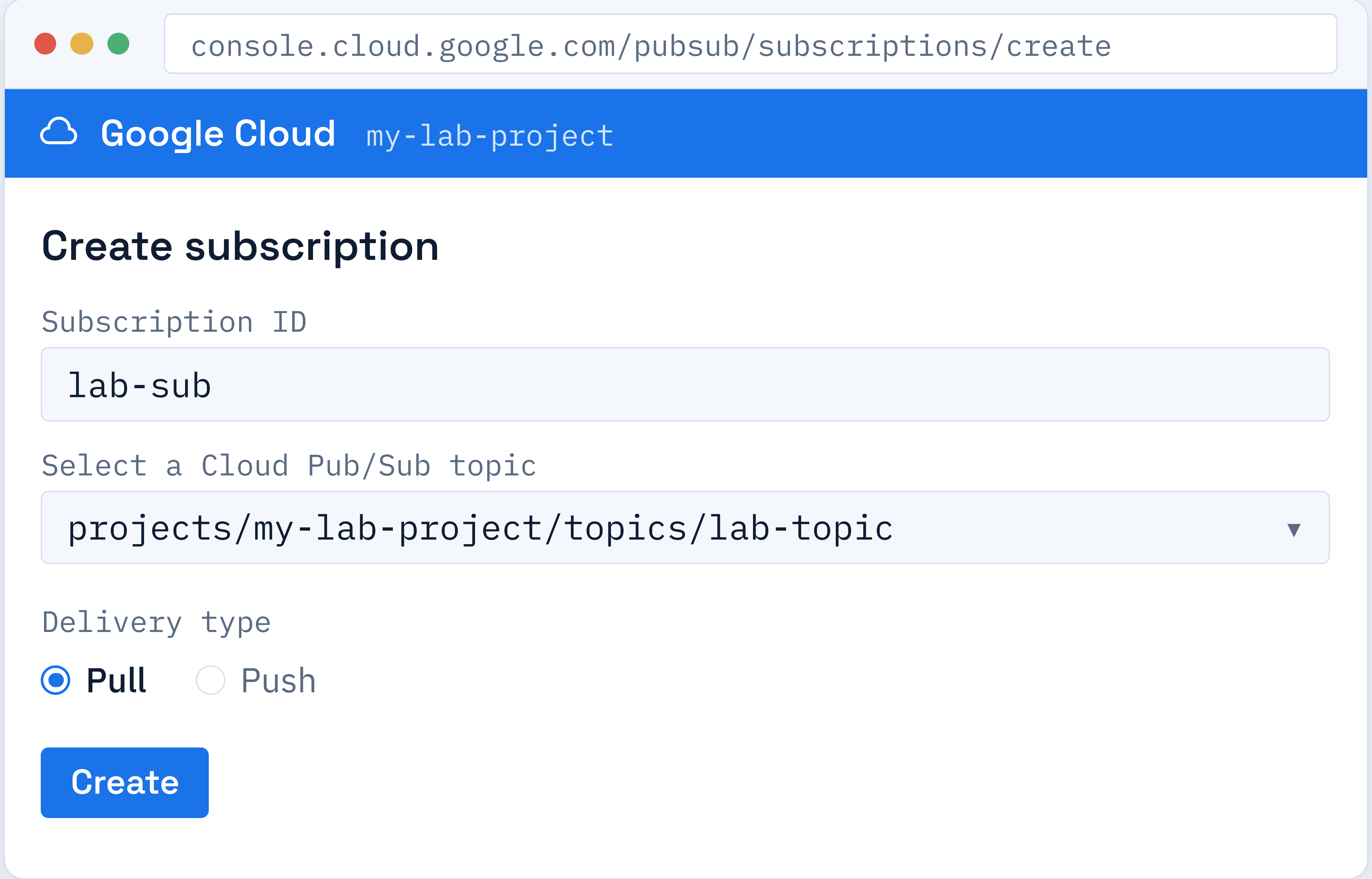
PUB/SUB LAB · STEP 2

Create a Subscription

STEPS

1. Open your topic and scroll to the **Subscriptions** section.
2. Click **Create Subscription**.
3. Choose **Pull** delivery type and click Create.

✓ **Expected result:** a pull subscription is attached to the topic.



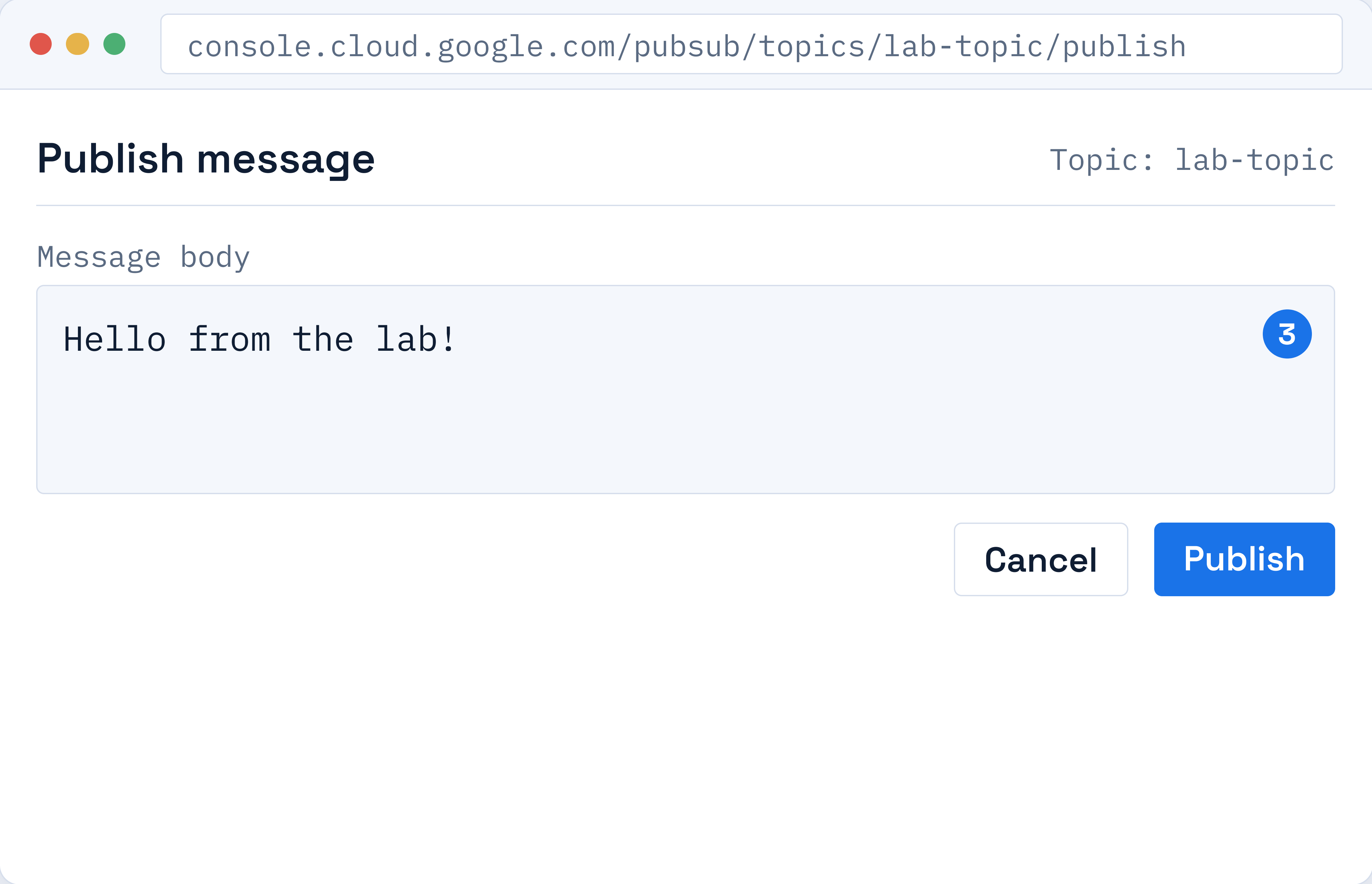
PUB/SUB LAB · STEP 3

Publish a Message

STEPS

1. Open the **lab-topic** and click the **Messages** tab.
2. Click **Publish Message**.
3. Type a message body and click **Publish**.

✓ **Expected result:** the message is accepted by the topic.

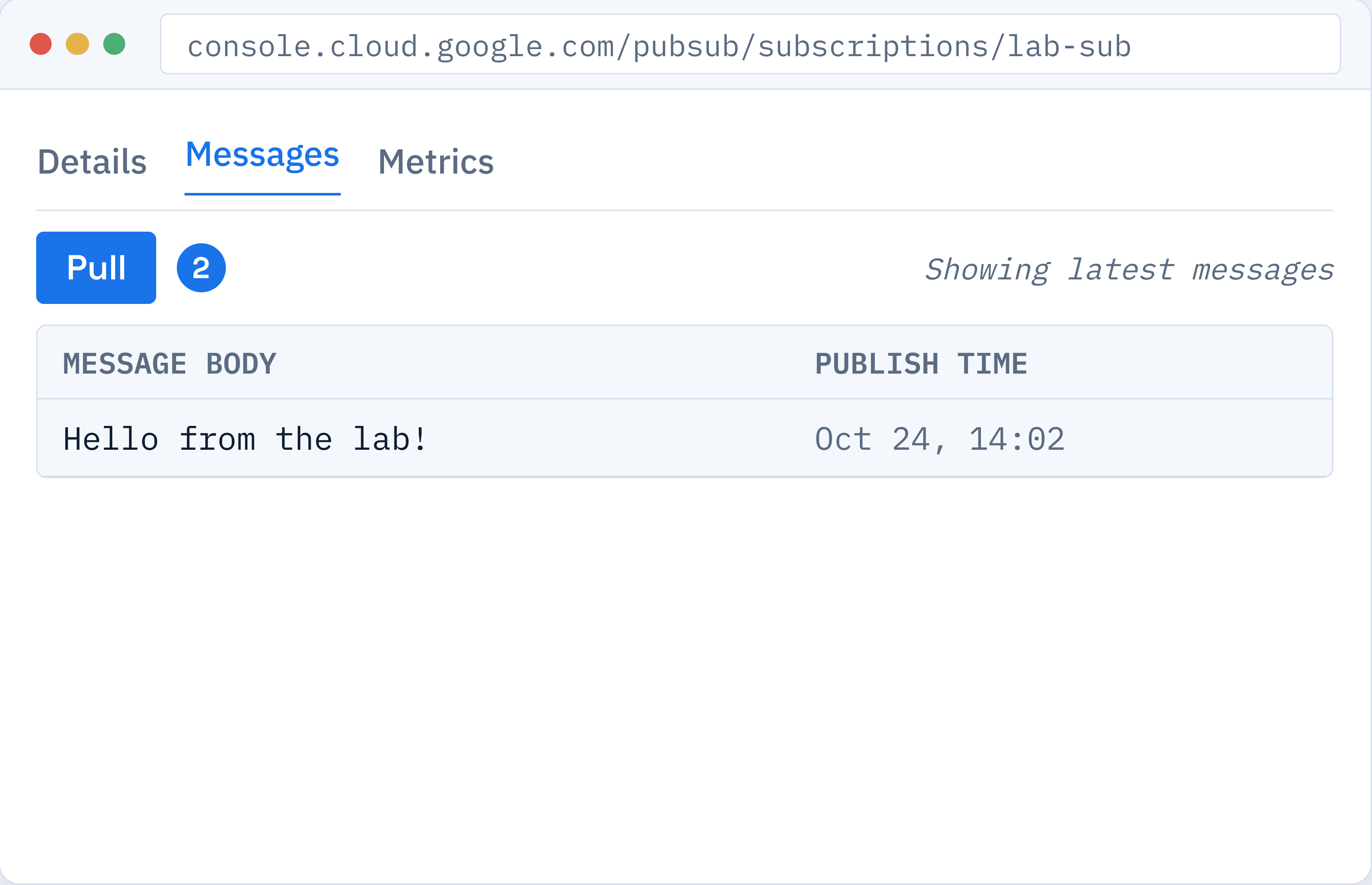


PUB/SUB LAB · STEP 4

Pull the Message

STEPS

1. Open the **lab-sub** subscription.
2. Click the **Pull** button in the Messages tab.
3. The message body appears in the results list.



✓ **Expected result:** the published message is shown in the pull results.

PUB/SUB LAB · STEP 5

Verify Delivery

STEPS

1. Confirm the pulled body matches what you published.
2. **Acknowledge** the message to remove it from the queue.
3. Check that the subscription has no backlog remaining.

✓ **Expected result:** the message body matches and the backlog is zero.

console.cloud.google.com/pubsub/subscriptions/lab-sub

Pull results

Unacked: 0

MESSAGE BODY	ACTION
Hello from the lab!	Ack 2

✓Message successfully acknowledged and removed from queue.

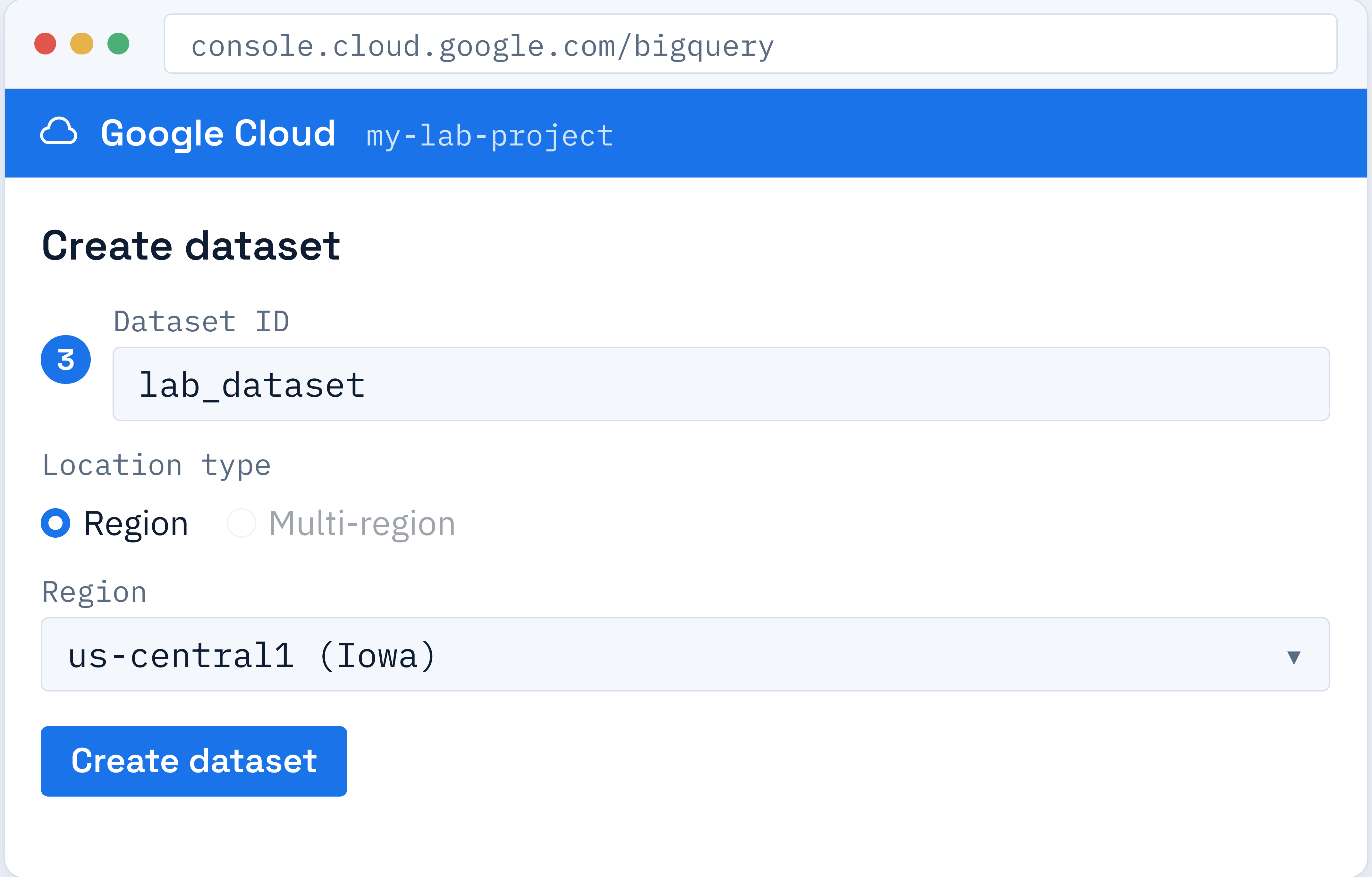
BIGQUERY LAB · STEP 1

Create a Dataset

STEPS

1. Open BigQuery.
2. Next to your project, click **Create Dataset**.
3. Enter a Dataset ID, e.g. `lab_dataset`.

✓ **Expected result:** the dataset appears under your project in the explorer.

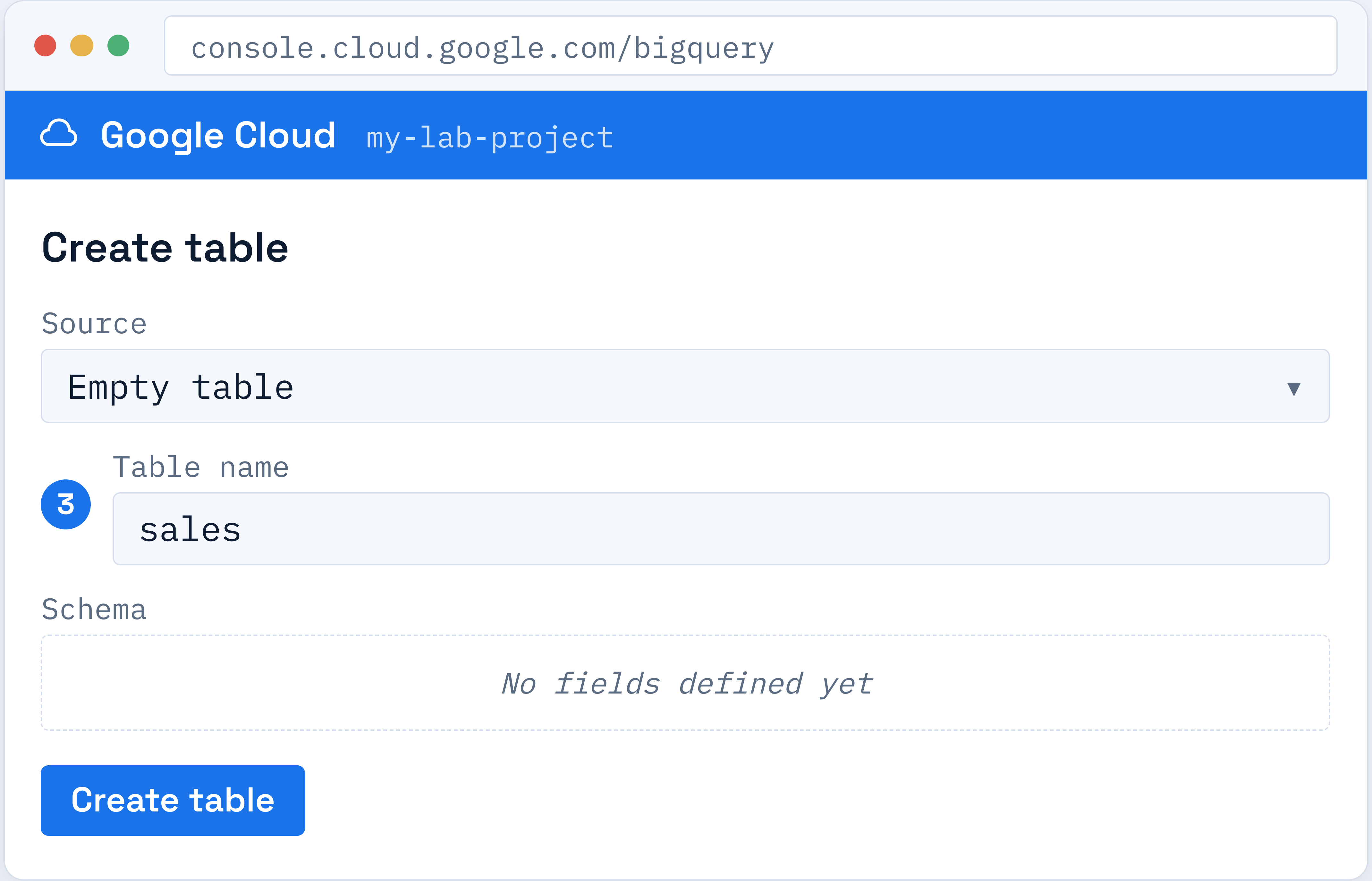


BIGQUERY LAB · STEP 2

Create a Table

STEPS

1. Open the dataset `lab_dataset`.
2. Click **Create Table**.
3. Name the table, e.g. `sales`.



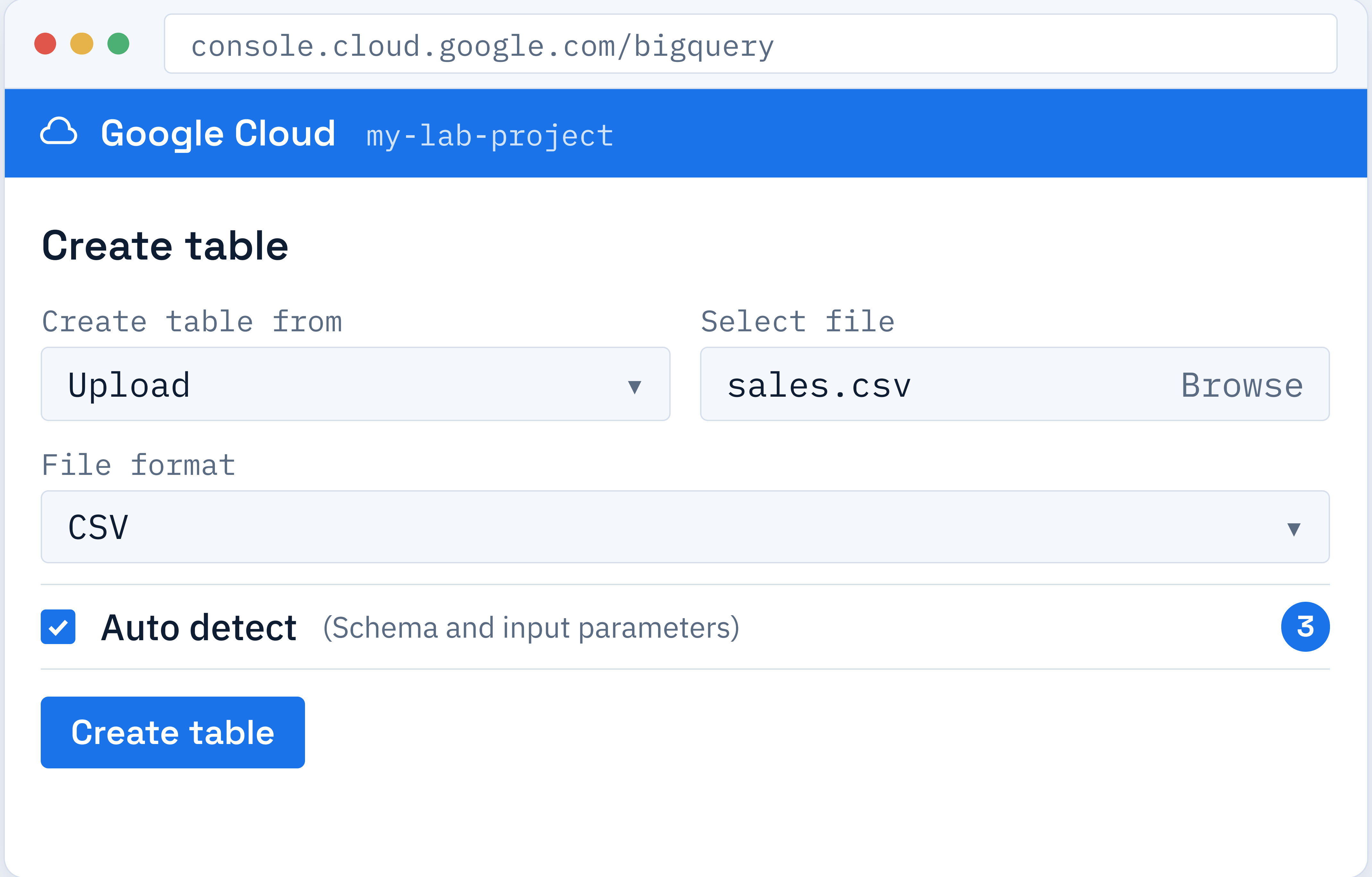
✓ **Expected result:** an empty table is created in the dataset.

BIGQUERY LAB · STEP 3

Load Data from a CSV

STEPS

1. In **Create Table**, set Source to **Upload**.
2. Choose a CSV file, e.g. `sales.csv`.
3. Enable **Auto-detect schema**.



✓ **Expected result:** the CSV loads and the table shows rows.

BIGQUERY LAB · STEP 4

Open the Query Editor

STEPS

1. Select the table `sales`.
2. Click **Query → In new tab**.
3. The SQL editor opens.

✓ **Expected result:** the query editor opens referencing your table.

BIGQUERY LAB · STEP 5

Run a SELECT Query

STEPS

1. Type a **SELECT** statement.
2. Click **Run** (the green button).
3. Wait for the job to finish.

✓ **Expected result:** the query runs and a results tab appears.

console.cloud.google.com/bigquery

Run

2

```
SELECT region, SUM(amount) AS total
FROM `lab_dataset.sales`
GROUP BY region
```

BIGQUERY LAB · STEP 6

Read the Results

STEPS

1. Open the **Query results** tab.
2. Review the returned rows in the table.
3. Note rows scanned and job execution time.

✓ **Expected result:** the results table shows aggregated rows.

console.cloud.google.com/bigquery

Query results | Job information

Row	region	total
1	us-central1	4,210
2	europe-west1	2,980
3	asia-east1	1,540

3 rows returned

Job completed in 0.4s

BIGQUERY LAB · COMMON ERRORS

Query Pitfalls

⚠ Table not found

Wrong dataset or table name provided in the query.

Fix: use `dataset.table` and check spelling.

⚠ Syntax error

Invalid SQL structure or missing clauses.

Fix: fix the clause BigQuery points to in the editor.

⚠ Schema mismatch

CSV columns don't match the target table types.

Fix: use auto-detect or manually fix the schema definition.

⚠ Permission denied

Missing BigQuery role to access data or run jobs.

Fix: grant BigQuery Data Viewer or Editor role.

WRAP-UP · CHALLENGE

Put It All Together

Combine what you built into one small pipeline.

1

HOST

Run a small web server on your VM and reach it through the firewall rule.

2

STORE

Upload its output file to your Cloud Storage bucket.

3

ANALYZE

Load that file into BigQuery and run a query over it.

No step-by-step here — use the labs as your reference.

WRAP-UP · MAPPING EXERCISE

Map the Architecture

In the sample below, identify which service family each box belongs to.

BOX A
VM Instance

→

BOX B
Message Topic

→

BOX C
Analytics Table

→

BOX D
Storage Bucket

- Compute
- Messaging
- Analytics
- Storage

WRAP-UP · SELF-CHECK

Final Lab Checklist

Confirm each lab is complete before you tear down resources.

- ✓ VM created and reachable over SSH
- ✓ Firewall rules allow HTTP and HTTPS
- ✓ Bucket created with a file uploaded
- ✓ Pub/Sub message published and pulled
- ✓ BigQuery query returned results
- All resources deleted to avoid charges

WRAP-UP · RECAP

Lecture by **Imran Sayyed**

5 Things You Mastered

You have built the core components of a cloud-native architecture.

- **Compute & Networking**
  Deploying VMs and securing them with VPC firewall rules.
- **Object Storage**
  Managing global buckets and unstructured data objects.
- **Asynchronous Messaging**
  Decoupling systems with Pub/Sub topics and subscriptions.
- **Data Analytics**
  Running SQL queries on massive datasets with BigQuery.
- **Security Best Practices**
  Applying least privilege and avoiding public resource exposure.